Using Euler's relation, this can be written in rectangular form as:

$$x[n] = \sum_{k=0}^{N-1} ReX[k]\left(\cos(2\pi kn/N) + j\sin(2\pi kn/N)\right) - \sum_{k=0}^{N-1} ImX[k]\left(\sin(2\pi kn/N) - j\cos(2\pi kn/N)\right)$$

EQUATION 31-8
The inverse complex DFT. This is Eq. 31-7 rewritten to show how each value in the frequency spectrum affects the time domain.

The compact form of Eq. 31-7 is how the inverse DFT is usually written, although the expanded version in Eq. 31-9 can be easier to understand. In words, each value in the real part of the frequency domain contributes a real cosine wave *and* an imaginary sine wave to the time domain. Likewise, each value in the imaginary part of the frequency domain contributes a real sine wave *and* an imaginary cosine wave. The time domain is found by adding all these real and imaginary sinusoids. The important concept is that each value in the frequency domain produces both a *real* sinusoid and an *imaginary* sinusoid in the time domain.

For example, imagine we want to reconstruct a unity amplitude cosine wave at a frequency of $2\pi k/N$. This requires a positive frequency and a negative frequency, both from the real part of the frequency spectrum. The two square markers in Fig. 31-1 are an example of this, with the frequency set at: $k/N = 0.23$. The positive frequency at 0.23 (labeled 1 in Fig. 31-1) contributes a cosine wave and an imaginary sine wave to the time domain:

$$\tfrac{1}{2}\cos(2\pi 0.23n) + \tfrac{1}{2}j\sin(2\pi 0.23n)$$

Likewise, the negative frequency at -0.23 (labeled 2 in Fig. 31-1) also contributes a cosine and an imaginary sine wave to the time domain:

$$\tfrac{1}{2}\cos(2\pi(-0.23)n) + \tfrac{1}{2}j\sin(2\pi(-0.23)n)$$

The negative sign within the cosine and sine terms can be eliminated by the relations: $\cos(-x) = \cos(x)$ and $\sin(-x) = -\sin(x)$. This allows the negative frequency's contribution to be rewritten:

$$\tfrac{1}{2}\cos(2\pi 0.23n) - \tfrac{1}{2}j\sin(2\pi 0.23n)$$